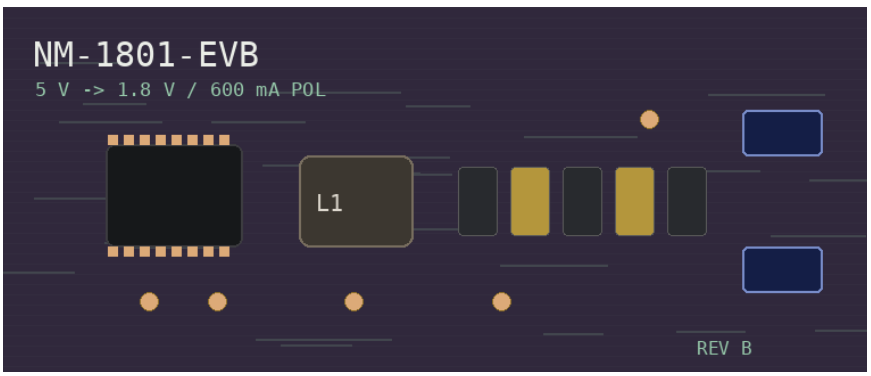
| 01 Understand | 02 Measure | 03 Infer |
|---|---|---|
| Read the context | Drive the bench | Diagnose the result |

| | |
|---|---|
| **Board** | Nimbus NM-1801-EVB |
| **Part number** | NM-1801 |
| **Vendor** | Nimbus Power |
| **Package** | QFN-12 |
| **Domain** | Power · Buck |
| **Scenario** | Marginal ripple |
| **Backend** | mock |
| **Bench** | Simulator |

**OVERALL: FAIL**

1 of 3 checks failed.




## Measurements

| TP | Check | Limit | Measured | Outcome |
|---|---|---|---|---|
| tp1 | Output voltage (500 mA load) | — | 1.7996 V | PASS |
| tp2 | Output voltage (no load) | — | 1.8027 V | PASS |
| tp3 | Output ripple (500 mA load) | — | 0.0281 V | FAIL |

## Diagnosis

**High output ripple, DC regulation clean — output capacitor.**

Output ripple measured **[TP 3: 0.028 V]** pp against a 12 mV limit (2.3× over spec).

DC output voltage is within spec at every load point **[TP 1: 1.800 V]**, **[TP 2: 1.803 V]**, so the pass device, the reference and the feedback loop are healthy. A ripple-only failure of this magnitude points to the output capacitor —

most likely an under-valued or high-ESR COUT, or poor placement near the regulator.

| | |
|---|---|
| **Root cause** | Output capacitor — high ESR or under-valued |
| **Next step** | Verify the COUT value and ESR against the datasheet's stability requirements, then re-test. |
| **Confidence** | 86% |

## Appendix A — SCPI log

### *tp1 — Output voltage (500 mA load)*

```
PSU → SOUR:VOLT 5.000
PSU → OUTP ON
LOAD → SOUR:CURR 0.500
LOAD → INP ON
DMM → CONF:VOLT:DC 10
DMM → MEAS:VOLT:DC?
DMM ⇐ 1.7996
```

### *tp2 — Output voltage (no load)*

```
PSU → SOUR:VOLT 5.000
PSU → OUTP ON
LOAD → SOUR:CURR 0.000
LOAD → INP ON
DMM → CONF:VOLT:DC 10
DMM → MEAS:VOLT:DC?
DMM ⇐ 1.8027
```

### *tp3 — Output ripple (500 mA load)*

```
PSU → SOUR:VOLT 5.000
PSU → OUTP ON
LOAD → SOUR:CURR 0.500
LOAD → INP ON
SCOPE → :CHAN1:COUP AC
SCOPE → :CHAN1:SCAL 0.02
SCOPE → :TIM:SCAL 1e-3
SCOPE → :MEAS:VPP? CHAN1
SCOPE ⇐ 0.02809
```